Jenifer Vincy A

Tirupur, Tamilnadu | jenifervincy@gmail.com | +91 7358197503 | Jenifer Vincy | jenifervincya

EDUCATION

KIT-KALAIGNARKARUNANIDHI INSTITUTE OF TECHNOLOGY 2024-2028
B.E. Electrical and Electronics Engineering Current CGPA: **8.79**
Coursework: Embedded C, Internet of Things (IoT), PCB Design, Microcontrollers

EXPERIENCE

Internship, iQuants Engineering Solutions Private Limited– Coimbatore, India (On-site) LINK | May-June 2026

- Studied ECU architectures and Software Defined Vehicle (SDV) concepts across body, powertrain, and chassis domains.
- Analyzed CAN, LIN, Ethernet, UART, and I2C protocols in embedded ECU workflows

Industrial Training, Sunshiv Electronic Solutions – Coimbatore, India (On-site) LINK | Dec 2025

- Designed PCB schematics and assembled circuits using Autodesk EAGLE
- Developed skills in component selection, circuit debugging, soldering, and PCB fault analysis

PUBLICATION

EDGE-AI Power Auditor: Decentralized Harmonic Analysis and NILM-Based Diagnostics (Article ID: ICOSAAS-429)

- Accepted at International Conference on Intelligent and Sustainable AI Systems - 2026, to appear in IEEE Xplore.
- Presented an STM32-based edge-AI system using NILM and harmonic analysis for real-time industrial fault detection.

PROJECTS

BoltOn.AI – Plug-and-Play Industrial IoT System for Smart Machine Monitoring and Predictive Maintenance LINK

- Built a real-time edge AI system for machine anomaly detection and predictive maintenance without manual inspection.
- Implemented a sensor data pipeline with live visualization, deployed on ESP32 with remote dashboard integration.

ResQPath – Intelligent & Adaptive Traffic Signal System LINK

- Built an RFID-based traffic signal system that prioritizes emergency vehicles to reduce response delays.
- Used IoT integration for real-time remote signal monitoring and adaptive, dynamic lane prioritization logic.

EDGE-AI Smart Auditor: Decentralized Harmonic Analysis and NILM-Based Diagnostics

- Built an STM32-based system for real-time power monitoring and load behavior classification.
- Applied NILM-based load classification and edge-based anomaly detection for energy diagnostics without cloud dependency.

TECHNICAL SKILLS

Programming: C, Embedded C (Proficient) | C++, Python, Java (Familiar) | DSA, OOPS
Technical Domains: PCB Design, ECU Architecture, Communication Protocols (CAN, LIN, Ethernet, UART, SPI, I2C), IIoT/IoT Systems
Microcontroller: Arduino, ESP32, STM32
Software Tools: Autodesk EAGLE, Git, GitHub, MATLAB (Basic)

CERTIFICATIONS & COURSES

- Technology Commercialization & New Product Development – NPTEL/IIT Bombay (Elite + Gold) LINK
- Google Prompting Essentials – Coursera/Google LINK
- Introduction and Programming with IoT Boards – Coursera/POSTECH LINK
- Python Foundation – Infosys Springboard LINK
- Git and GitHub – CodeChef LINK

COMPETITIVE PROGRAMMING

Codechef: Max Rating – **1170** | DSA Max Rating – **1167** | Badges – **3** | Solved problems – **650+** | Position – **1 Star**
Leetcode: Max Rating **– 1605** | Badges – **4** | Solved Problems – **180+**

ACHIEVEMENTS

- **TECHNOXIAN 9.0 World Robotics Championship, Delhi –** Participated in RC Car Racing
- **HackSmart, 36 hrs Hackathon –** Participated and successfully completed our prototype "Waste to Watts"
- **TNWISE 2025 by TANCAM, Hackathon for Women –** Participated and presented my idea on Women's Safety Gadget
- **TN-IMPACT 2026 by TANCAM –**Provide solution for the Problem Statement Number-TNI26022